

Tirth Laheri

+1 (615)-497-1076 | tirthlaheri194@gmail.com | linkedin.com/in/tirthlaheri/ | github.com/Tirth1411

EDUCATION

Middle Tennessee State University

Murfreesboro, TN

Master of Science in Data Science GPA: 4.00/4.00

Aug 2024 – Present

- Relevant Coursework: Machine Learning, Statistical Methods, Data Mining, Data Visualization

Parul University

India

Bachelor of Technology in Computer Science (IoT) GPA: 9.06/10.00

Jul 2020 – May 2024

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Cloud Computing

EXPERIENCE

Graduate Assistant (Administrative)

Aug 2025 – Present

Physician Assistant Department, Middle Tennessee State University

Murfreesboro, TN

- Led the design and rollout of a **SQL-driven data pipeline** to extract, transform, and automate real-time pass-rate and progression metrics for cohorts of 15–20 students, cutting reporting time by **~90%**.
- Wrote SQL queries and built ETL and data validation workflows to aggregate attendance, pass-rate, and progression data across 3+ cohorts, training department staff on the new system.
- Presented data-quality recommendations to faculty leadership, reducing administrative processing time by **~40%**.

Data Science Intern

Dec 2023 – Mar 2024

SlashMark

Hyderabad, IN

- Compared **4 classification models** in **Python** and **Scikit-learn** on a **10,000+ record** dataset, conducting EDA and feature engineering across **20+ variables** to identify **5 key predictors** and improve F1-score by **12%** over the baseline.
- Delivered **weekly analytical reports** and visualizations in **Matplotlib** and **Seaborn** to a 4-person data science team, informing iteration on model selection and feature strategy.

PROJECTS

Multilingual Legal Document Intelligence | *LangChain, OpenAI, pgvector, FastAPI*

Jan 2026 – Present

- Developed a multilingual legal-intelligence platform with a **RAG** architecture (LangChain, OpenAI GPT, PostgreSQL pgvector) enabling semantic search and Q&A across 100+ legal PDF documents in 5+ languages.
- Integrated **PaddleOCR** for layout-aware document parsing and exposed the pipeline through scalable **FastAPI** services, achieving 95%+ OCR accuracy and 50–500ms retrieval latency.

Advanced Agentic RAG System | *Python, LangChain, Multi-Agent, RAG, LLM, Streamlit*

Mar 2024

- Engineered a stateful **10-agent RAG workflow** (Router, Retriever, Grader, Reformulator, WebSearcher, Synthesizer, Generator, FactChecker, SafetyChecker, Clarifier) with a self-correcting retrieval loop that triggers reformulation or live web search via the Tavily API on low-confidence results.
- Backed the pipeline with a **ChromaDB** ingestion layer (PDF/DOCX/TXT/URL) and a configurable **Streamlit** interface for end-to-end querying.

Diabetic Retinopathy Screening | *Python, TensorFlow, Image Processing, Deep Learning*

Feb 2024

- Built a diabetic retinopathy screening pipeline combining **CNNs** and **RNNs** for automated image classification, achieving **~95% diagnostic accuracy** through image preprocessing and feature extraction while maintaining the sensitivity and specificity needed for early detection.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++, R, JavaScript, HTML/CSS

ML, Deep Learning & GenAI: Machine Learning, Deep Learning, Statistical Analysis, Computer Vision, NLP, EDA, Feature Engineering, LangChain, OpenAI API, Groq, RAG, Prompt Engineering, Vector Databases (ChromaDB, pgvector)

Frameworks, Tools & Platforms: TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, Streamlit, Git, VS Code, Google Colaboratory, PostgreSQL, Power BI, Tableau, Cloud Computing

PUBLICATIONS

Visible Light Communication Based Indoor Positioning System

2024

IEEE Xplore Digital Library

- Collaborated with a **4-member research team** to design and develop a VLC-based indoor localization system using photodiode sensors and microcontrollers, improving positioning precision through light-intensity calibration and ML-driven error correction.
- *Published in:* IEEE Xplore Link – ieeexplore.ieee.org/document/10601381